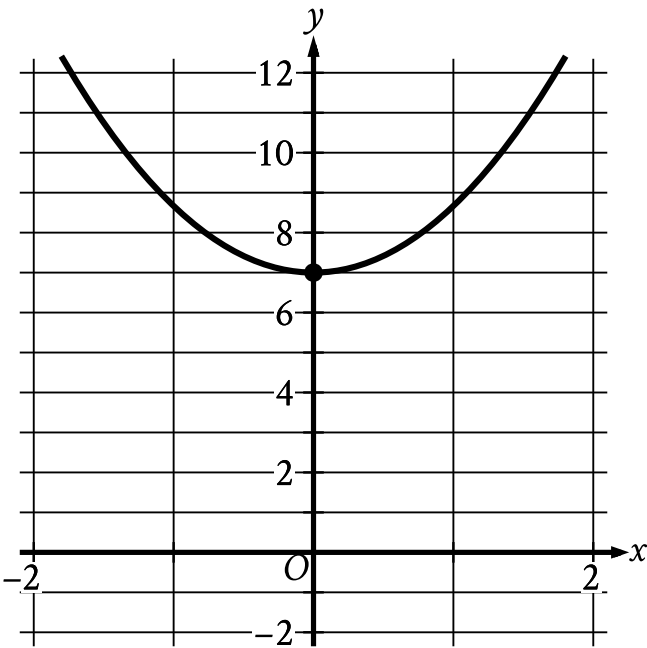


Question ID 2f4eafcc

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	

ID: 2f4eafcc

PSAT2 M 3.1



The parabola shown intersects the y -axis at the point (x, y) . What is the value of y ?

Question ID 2e379126

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: 2e379126

PSAT2 M 3.2

The function g is defined by $g(x) = 4x - 6$. What is the value of $g(-7)$?

- A. ~~-3~~4
- B. ~~-2~~2
- C. $-\frac{13}{4}$
- D. $-\frac{1}{4}$

Question ID e006209c

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear inequalities in one or two variables	

ID: e006209c

PSAT2 M 3.3

A geologist needs to collect at least **67** samples of lava from a volcano. If the geologist has already collected **63** samples from the volcano, what is the minimum number of additional samples the geologist needs to collect?

- A. **130**
- B. **63**
- C. **4**
- D. **0**

Question ID 0d43db90

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	

ID: 0d43db90

PSAT2 M 3.4

The perimeter of triangle ABC is **17** inches, the length of side AB is **4** inches, and the length of side AC is **7** inches. What is the length, in inches, of side BC ?

- A. **4**
- B. **6**
- C. **7**
- D. **11**

Question ID a130fcdc

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: a130fcdc

PSAT2 M 3.5

$g(x) = 11x + 4$

For the given linear function g , which table shows three values of x and their corresponding values of $g(x)$?

A.

x	$g(x)$
-1	7
0	11
1	15

B.

x	$g(x)$
-1	-4
0	0
1	4

C.

x	$g(x)$
-1	-7
0	4
1	15

D.

x	$g(x)$
-1	-11
0	0
1	11

Question ID c4259674

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	

ID: c4259674

PSAT2 M 3.6

The function f is defined by $f(x) = 4x^{-1}$. What is the value of $f(21)$?

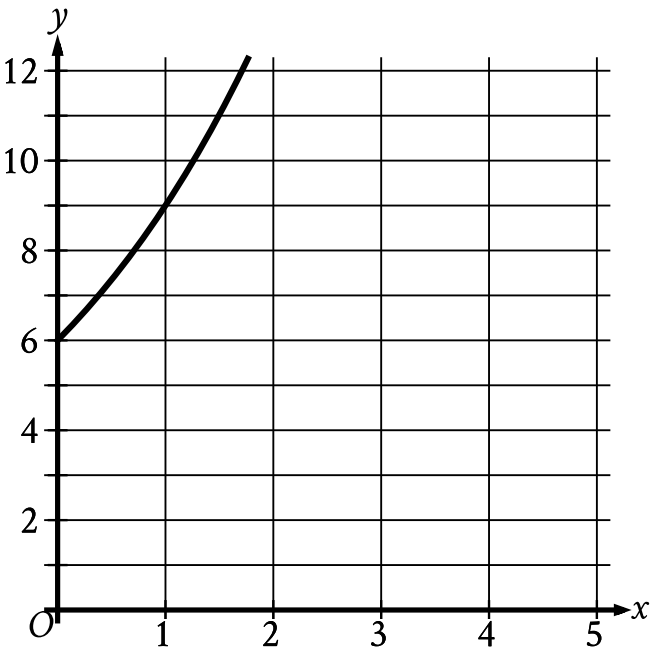
- A. ~~-84~~
- B. $\frac{1}{84}$
- C. $\frac{4}{21}$
- D. $\frac{21}{4}$

Question ID f1fa0821

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	

ID: f1fa0821

PSAT2 M 3.7



The graph gives the estimated population y , in thousands, of a town x years since 2003, where $0 \leq x \leq 5$. Which of the following best describes the increase in the estimated population from $x = 0$ to $x = 1$?

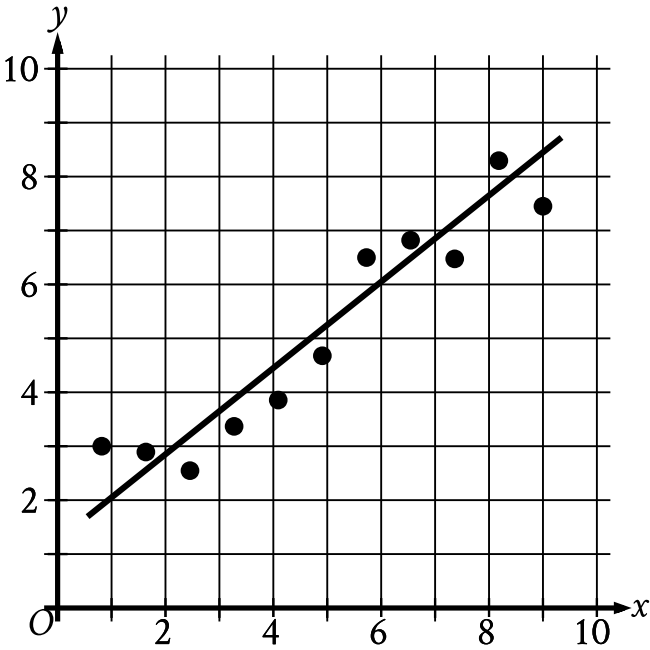
- A. The estimated population at $x = 1$ is 0.5 times the estimated population at $x = 0$.
- B. The estimated population at $x = 1$ is 1.5 times the estimated population at $x = 0$.
- C. The estimated population at $x = 1$ is 2.5 times the estimated population at $x = 0$.
- D. The estimated population at $x = 1$ is 3.5 times the estimated population at $x = 0$.

Question ID ad7dbb22

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Two-variable data: Models and scatterplots	

ID: ad7dbb22

PSAT2 M 3.8



The scatterplot shows the relationship between two variables, x and y . A line of best fit is also shown. For how many of the 11 data points does the line of best fit predict a greater y -value than the actual y -value?

Question ID 55447be2

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	

ID: 55447be2

PSAT2 M 3.9

$$y = \frac{1}{3}x - 14$$
$$y = -x + 18$$

The solution to the given system of equations is (x, y) . What is the value of x ?

Question ID 8ad64841

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: 8ad64841

PSAT2 M 3.10

A linear function f gives a company’s profit, in dollars, for selling x items. The company’s profit is \$220 when it sells 8 items, and its profit is \$320 when it sells 10 items. Which equation defines f ?

- A. $f(x) = 150x - 320$
- B. $f(x) = 32x$
- C. $f(x) = 50x - 10x$
- D. $f(x) = 50x - 180$

Question ID 40049d49

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in one variable	

ID: 40049d49

PSAT2 M 3.11

$4x + 12 = \frac{a(x+3)}{2}$

In the given equation, a is a constant. If the equation has infinitely many solutions, what is the value of a ?

- A. 0
- B. 3
- C. 8
- D. 12

Question ID d72a2b4d

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Percentages	

ID: d72a2b4d

PSAT2 M 3.12

In March, the price of a collectible card was **\$15.50**. In April, the price of the collectible card was **\$17.36**. The price of the collectible card in April was *p*% of the price of the collectible card in March. What is the value of *p*?

- A. 12
- B. 88
- C. 112
- D. 188

Question ID 3148fe3e

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: 3148fe3e

PSAT2 M 3.13

$x^2 + y + 10 = 10$ $8x + 16 - y = 0$ The solution to the given system of equations is (x, y) . What is the value of x ?

- A. -16
- B. -4
- C. 2
- D. 8

Question ID be1b8c74

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: be1b8c74

PSAT2 M 3.14

$x = 8a(b + 9)$

The given equation relates the positive numbers a , b , and x . Which equation correctly expresses a in terms of b and x ?

- A. $a = \frac{x}{8} - (b + 9)$
- B. $a = \frac{x}{8(b+9)}$
- C. $a = \frac{8(b+9)}{x}$
- D. $a = 8x(b + 9)$

Question ID fde6f3bb

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Equivalent expressions	

ID: fde6f3bb

PSAT2 M 3.15

$g(x) = \frac{3}{5}x + \frac{7}{6}$ $h(x) = 6x - 5$

The functions g and h are defined by the equations shown. Which expression is equivalent to $g(x) \cdot h(x)$?

- A. $\frac{18x^2}{5} - \frac{35}{6}$
- B. $\frac{18x^2}{5} + \frac{27x}{11} - \frac{35}{6}$
- C. $\frac{18x^2}{5} - 4x - \frac{35}{6}$
- D. $\frac{18x^2}{5} + 4x - \frac{35}{6}$

Question ID 301faf80

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	

ID: 301faf80

PSAT2 M 3.16

The product of two positive integers is ~~4~~**62**. If the first integer is **5** greater than twice the second integer, what is the smaller of the two integers?

Question ID 40491607

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	

ID: 40491607

PSAT2 M 3.17

$f(x) = (x - 1)(x + 3)(x - 2)$

In the xy -plane, when the graph of the function f , where $y = f(x)$, is shifted up 6 units, the resulting graph is defined by the function g . If the graph of $y = g(x)$ crosses through the point $(4, b)$, where b is a constant, what is the value of b ?

Question ID 50b99b2d

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	■ ■ ■

ID: 50b99b2d

PSAT2 M 3.18

Objects R and S each travel at a constant speed. The speed of object R is half the speed of object S. Object R travels a distance of $4x$ inches in y seconds. Which expression represents the time, in seconds, it takes object S to travel a distance of $24x$ inches?

- A. $12y$
- B. $3y$
- C. $16y$
- D. $6y$

Question ID 7355b9d9

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Equivalent expressions	■ ■ ■

ID: 7355b9d9

PSAT2 M 3.19

If $k - x$ is a factor of the expression $-x^2 + \frac{1}{29}nk^2$, where n and k are constants and $k > 0$, what is the value of n ?

- A. -29
- B. $-\frac{1}{29}$
- C. $\frac{1}{29}$
- D. 29

Question ID dcf63c94

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	

ID: dcf63c94

PSAT2 M 3.20

$f(x) = 272(2)^x$

The function f is defined by the given equation. If $h(x) = f(x - 4)$, which of the following equations defines function h ?

- A. $h(x) = 17(2)^x$
- B. $h(x) = 68(2)^x$
- C. $h(x) = 272(16)^x$
- D. $h(x) = 272(8)^x$

Question ID f731d88b

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	■ ■ ■

ID: f731d88b

PSAT2 M 3.21

In convex pentagon $ABCDE$, segment AB is parallel to segment DE . The measure of angle B is 139 degrees, and the measure of angle D is 174 degrees. What is the measure, in degrees, of angle C ?

Question ID 2c288148

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: 2c288148

PSAT2 M 3.22

$\sqrt{k - x} = 58 - x$

In the given equation, k is a constant. The equation has exactly one real solution. What is the minimum possible value of $4k$?